Year 9 5.2 Yearly Exam Revision

Booklet A

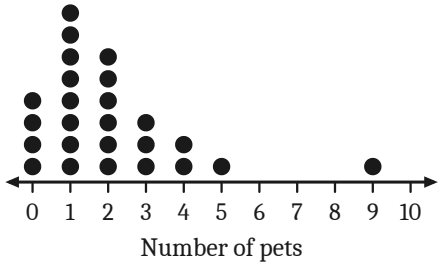
Answer every question in your **maths workbook**.
Show all working. A calculator is allowed.

Section 1: Data

1 Classify each type of data as **categorical** or **numerical**. If it is numerical, state whether it is **discrete** or **continuous**.

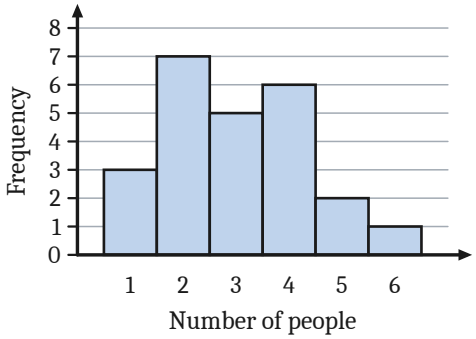
a the number of pets in a household
b the colours of students' school bags
c the mass of a newborn baby
d the number of text messages sent in a day
e the time taken to walk to school
f clothing sizes (S, M, L)

2 The dot plot shows the number of pets owned by each student in a class.



a How many students are in the class?
b Describe the shape of the distribution (symmetrical, positively skewed or negatively skewed).
c Where do the scores cluster?
d Is there an outlier? If so, what is it?
e Find the mode.

3 The histogram shows the number of people living in each house in a street.



a How many houses are in the street?
b What is the mode?
c What is the range?
d How many houses have more than 3 people?

4 A basketball player scored these points in 8 games:
12, 15, 9, 14, 41, 13, 11, 13

a Which score is an outlier?
b Find the mean and the median of all 8 scores.
c Find the mean and the median without the outlier.
d Which measure of centre is changed more by the outlier? Which one better describes the data?

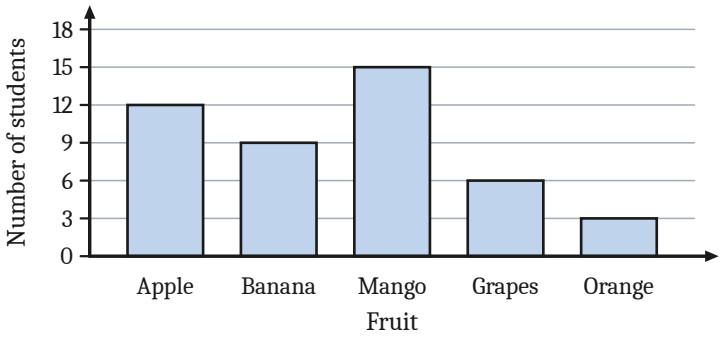
5 The stem-and-leaf plot shows the marks (out of 50) of 12 students in a test.

Stem	Leaf
2	3 7
3	1 2 5 5 8
4	0 2 4 6 7

Key: 3 | 5 = 35 marks

a Find the mean, correct to 1 decimal place.
b Find the median.
c Find the mode.
d Find the range.

6 Students in Year 9 were asked to name their favourite fruit. The results are shown in the column graph.



a Which fruit was the most popular?
b How many more students chose mango than grapes?
c How many students were surveyed?
d What percentage of the students chose apple or banana? Answer correct to 1 decimal place.

7 The back-to-back stem-and-leaf plot shows the test marks (out of 50) of two Year 9 classes.

a Find the median mark for each class.

b Find the range for each class.

c Which class did better in the test? Give a reason.

d Which class has the more consistent marks? Explain.

9X	Stem	9Y
	1	4 8
8 5 1	2	6 9
8 6 5 3 2 0	3	7
1	4	0 2 5 7 9

Key: 3 | 2 = 32 marks

Section 2: Surface area and volume

Formulas
1 kL = 1000 L, 1 L = 1000 mL, 1 L = 1000 cm³, 1 m³ = 1000 L
Parallelogram $A = bh$; kite and rhombus $A = \frac{1}{2}xy$; trapezium $A = \frac{1}{2}h(a + b)$
Circle $C = 2\pi r$, $A = \pi r^2$; cylinder $V = \pi r^2 h$, $SA = 2\pi r^2 + 2\pi rh$; rectangular prism $V = lwh$

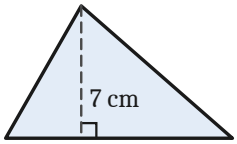
8 Convert:

a 2.4 kL to L **b** 3750 mL to L **c** 0.6 L to mL

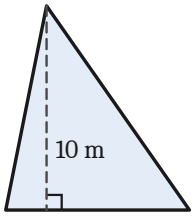
d 8200 L to kL **e** 4000 cm³ to L **f** 2.5 m³ to L

9 Which formula gives the area of a triangle: $A = bh$, $A = \frac{1}{2}bh$, $A = 2bh$ or $A = \frac{1}{2}(b + h)$? Use it to find the area of each triangle.

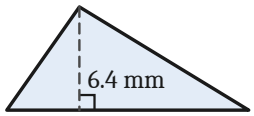
a



b

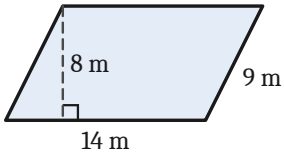


c

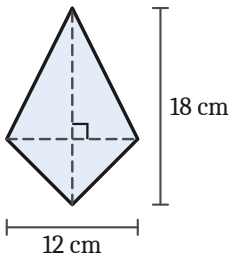


10 Find the area of each shape.

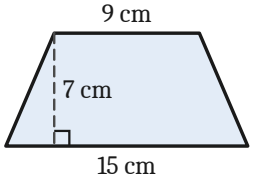
a



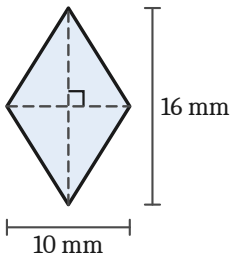
b



c

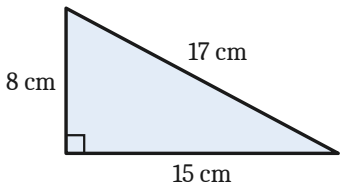


d

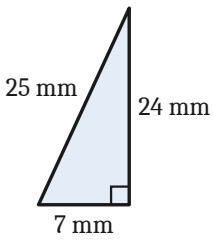


11 For each right-angled triangle, find (i) the perimeter and (ii) the area.

a

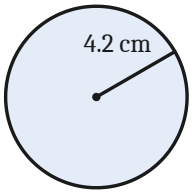


b

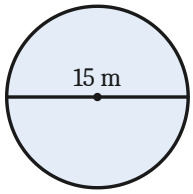


12 For each circle, find (i) the circumference and (ii) the area, correct to 1 decimal place.

a



b

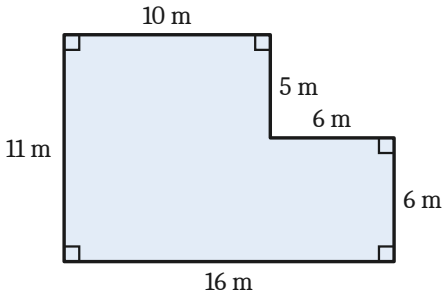


13 The plan shows a paved courtyard. All of the corners are right angles.

a Find the perimeter of the courtyard.

b Find the area of the courtyard.

c Pavers cost $38.50 per square metre. Find the cost of paving the whole courtyard.

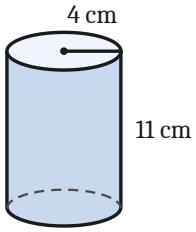


14 A closed can of soup is a cylinder with a radius of 4 cm and a height of 11 cm.

a Find the volume of the can, correct to 1 decimal place.

b How many millilitres does the can hold? (1 cm^3 = 1 mL)

c Find the surface area of the can, correct to 1 decimal place.

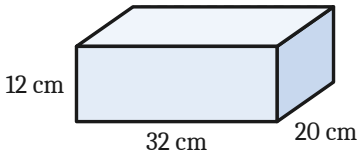


15 A closed shoe box is a rectangular prism 32 cm long, 20 cm wide and 12 cm high.

a Find the volume of the box.

b How many litres does the box hold?

c Find the surface area of the box.



Section 3: Mixed topics

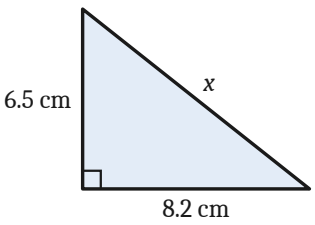
Formulas
Simple interest $I = Prn$; percentage profit = profit ÷ cost price × 100%; 1 year = 52 weeks = 26 fortnights = 12 months
Pythagoras' theorem $c^2 = a^2 + b^2$; $\sin\theta$ = opp ÷ hyp, $\cos\theta$ = adj ÷ hyp, $\tan\theta$ = opp ÷ adj

16 If $p = 6$ and $q = 3$, evaluate:

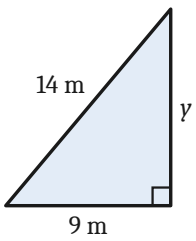
a $2p + q$ **b** $p^2 - q$ **c** $3q^2$ **d** $(p - q)^2$

17 Find the value of each pronumeral, correct to 1 decimal place.

a



b



18 Solve each problem.

a A jacket marked at $85 is discounted by 20%. Find the sale price.

b A $240 bike is discounted by 15%. How much is saved, and what is the sale price?

c A shop buys a TV for $650 and sells it for $799. Find the profit, and the profit as a percentage of the cost price (to 1 decimal place).

19 Simplify, writing each answer in index notation.

a $3x^4 \times 5x^3$ **b** $12y^9 \div 4y^3$ **c** $(m^2)^5$ **d** $2a^3b \times 6ab^2$

20 Harper is an engineer. She earns an annual salary of $78 260. How much is she paid:

a per week? **b** per fortnight? **c** per month?

21 Simplify:

a $7x + 2y - 3x + 5y$ **b** $6m \times 4n$ **c** $20ab \div 5a$ **d** $9k - 4 + 2k + 10$

22 Expand and simplify:

a $4(3x - 2)$ **b** $-5(2a + 3)$

c $2(x + 7) + 3(x - 4)$ **d** $5(2y - 1) - 3(y - 6)$

23 Factorise:

a $8x + 20$ **b** $15m - 10$ **c** $a^2 + 6a$ **d** $12y^2 - 18y$

24 Find the simple interest earned on each investment.

a \$5400 at 3.5% p.a. for 4 years **b** \$12 000 at 5.2% p.a. for 30 months

25 Solve each equation.

a $4x + 9 = 37$ **b** $x \div 3 - 5 = 2$ **c** $6m - 7 = 2m + 25$

d $3(2y - 1) = 4y + 15$ **e** $5k - 3(k + 2) = k + 4$

26 Priya sells houses. She is paid a retainer of \$650 per week plus 1.8% commission on her sales.

a Last week Priya sold a house for \$85 000. Find her total pay for the week.

b Tom earns 6% commission only (no retainer). Find his pay for a week in which he sells \$23 400 of goods.

27 Leo works in a café. He is paid \$28.40 per hour for the first 38 hours each week and time-and-a-half for each hour after that. Last week he worked 45 hours.

a Find the pay for the first 38 hours.

b Find the pay for the overtime hours.

c Find the total pay for the week.

28 A plumber charges a call-out fee of \$90 plus \$65 per hour. The cost, C dollars, for h hours is $C = 90 + 65h$.

a Find the cost of a 3-hour job.

b Another job cost \$415. Write an equation and solve it to find how many hours the job took.

29 A ladder 4.8 m long leans against a vertical wall. The foot of the ladder is 1.3 m from the wall.

a How high up the wall does the ladder reach? Answer correct to 2 decimal places.

b Find the angle θ the ladder makes with the ground, correct to the nearest degree.

